

Oracle Memory Structures

Purpose

- o Utilizes server's physical memory to manage various aspects of an Oracle Instance.

Components held

- o Oracle executable code.
- o Session information
- o Individual processes linked with the database.
- o Shared information between processes, like database object locks.
- o User and data dictionary SQL statements
- o Cached data eventually stored on disk, such as data blocks and transaction information.

SGA:

- o Allocated data area for an Oracle instance, containing various memory structures.

Software code area:

- o Houses Oracle executables.

PGA:

- Private area for each server and background process, with one PGA allocated per process.

SGA: System Global Area

⇒ SGA is a shared memory region that holds critical data and control information for the Oracle instance.

⇒ It's shared among all users of the database instance.

⇒ When an Oracle instance is started, memory is allocated for the SGA based on the values specified in the initialization parameter file or hard coded in the Oracle Software.

⇒ Many of the parameters that control the size of the various parts of the SGA are dynamic.

The size of SGA is determined by initialization parameters like SGA_MAX_SIZE and SGA_TARGET.

⇒ Components within SGA include:

- Buffer Cache
- Shared Pool
 - ↳ Library cache
 - ↳ data dictionary cache
- Redo log buffers
- Large pool
- Java pool
- Streams Pool

Granules:-

⇒ The smallest amount of memory that can be added and subtracted from the SGA.

⇒ Memory in the SGA is allocated in units of granules.

⇒ A granule can be either 4MB or 16MB depending on the size of the SGA.

⇒ If the SGA is less than or equal to 128MB, a granule is 4MB, otherwise it is 16MB.

DBuffer Caches:

⇒ Stores data blocks from disk that are recently read to fulfill select statements or contain modified blocks from DML statements. Uses LRU algorithm.

Dynamic allocation:

• Memory area in the SGA holding these blocks is dynamic since Oracle9i.

Dirty buffers:

• Also known as dirty buffers because contains new blocks or old blocks.

Multiple block sizes:

• Supports tablespaces with up to five different block sizes, each requiring its own buffers cache.

SGA: System Global Area

⇒ SGA is a shared memory region that holds critical data and control information for the Oracle instance.

⇒ It's shared among all users of the database instance.

⇒ When an Oracle instance is started, memory is allocated for the SGA based on the values specified in the initialization parameter file or hard-coded in the Oracle Software.

⇒ Many of the parameters that control the size of the various parts of the SGA are dynamic.

⇒ The size of SGA is determined by initialization parameters like SGA_MAX_SIZE and SGA_TARGET.

⇒ Components within SGA include:

- Buffer caches
- Shared Pool
 - Library cache
 - data dictionary cache
- Redo log buffers
- Large pool
- Java pool
- Streams Pool

Granules:-

→ The smallest amount of memory that can be added and subtracted from the SGA.

⇒ Memory in the SGA is allocated in units of granules.

⇒ A granule can be either 4MB or 16MB depending on the size of the SGA.

⇒ If the SGA is less than or equal to 128MB, a granule is 4MB, otherwise it is 16MB.

Buffer Caches

⇒ Stores data blocks from disk that are recently read to fulfill select statements or contain modified blocks from DML statements. Uses LRU algorithm.

Dynamic allocation

• Memory area in the SGA holding these blocks is dynamic since Oracle9i.

Dirty buffers

• Also known as dirty buffers because contains new blocks or old blocks.

Multiple block sizes

• Supports tablespaces with up to five different block sizes, each servicing its own buffer cache.

Adaptability:

- Allows adjusting `DB_CACHE_SIZE` and `DB_nk_CACH_SIZE` dynamically to accommodate varying processing and transactional needs without instance restart.

Additional caches:

- Oracle supports two additional caches with the same block size as the default:

- Keep buffer pool
- Recycle buffer pool

- These pools allocate memory independently of other caches in the SGA.

Table Placement:

- Tables data blocks can be assigned to specific pools using the `BUFFER_POOL KEEP` or `BUFFER_POOL RECYCLE` clause in the `STORAGE` clause.

Performance Optimizations:

- Placing frequently used tables in the `KEEP` buffer pool minimizes I/O, enhancing overall performance.

Shared Pools:

- Stores commonly used SQL statements and PL/SQL code.

Components:

Library cache:

- Stores passed SQL and PL/SQL statements along with their execution plans.

Data dictionary cache:

- Holds metadata about the database structures.

Sizing:

- Controlled by the `SHARED_POOL_SIZE` initialization parameter.

Dynamic resizing:

- Can be resized dynamically as long as the total SGA size does not exceed `SGA_MAX_SIZE` or `SGA_TARGET`.

reduces Overhead:

- Minimizes the need for reparsing, enhancing performance.

(*) Library cache:

functions:

- Stores information about SQL and PL/SQL statements executed.

against the database.

Shared Resources:

- Shared among all users, allowing multiple users to potentially reuse the same SQL statement.

Contents:

- Contains the SQL statement, its execution plan and parse tree.

Performance Optimizations:

- Reusing execution plans and parse trees for identical SQL statements improves execution time.

Impact of size:

- If the library cache is too small, execution plans and parse trees are flushed out, necessitating frequent reloads of SQL statements.

(★) Data dictionary cache:

Purposes:

- Stores metadata about the database, its structures, privileges and roles.

Ownership:

- Tables are owned by SYS and SYSTEM schemas.

Contents:

- Contains essential information for database operation.

Cache function:

- Holds a subset of data dictionary table columns after reading into the buffer cache.

Continuous Use:

- Data blocks from these tables are regularly accessed to process user queries and DML commands.

Impact of size:

- If too small, requests for data dictionary information lead to additional I/O operations.

Recursive calls:

- Insufficient cache size results in I/O bound data dictionary requests, known as recursive calls, which should be avoided by correctly sizing the cache.

(III) Large pool:

Purpose:

- Optional area within the SGA.

Functionality:

- Used for transactions involving multiple databases, manage buffers for parallel queries, and RMAN parallel backup and restore operation.

Memory Allocation:

- Provides large blocks of memory for operations requiring substantial memory allocation.

Dynamic Sizing:

- Controlled by the initialization parameter LARGE_POOL_SIZE.

(IV) Java Pool:

Def: ◦ optional segment within java code.

Purpose: ◦ Utilized by the Oracle JVM for java code and data within a user session.

Function:

- Stores java-related data structures like classes and objects, similar how to SQL and PL/SQL are cached in the share pool. Provides memory allocation for java objects & java execution.

Analogy:

- Dedicated to JVM and related components.

(V) Redo Log buffers: - Circular buffers

Function:

- Stores recent changes made to data blocks in datafiles.

Write Triggers:

- When one-third full.
- Every three seconds.
- From Oracle Database log, when 1MB of redo is accumulated.

Importance:

- Critical for database recovery in case of instance crash before data blocks are written from buffer-cache to data-files. Each Oracle database instance has its own redo log.

size LOG-BUFFER sets the size in the bytes. The default value of this size is four times the DB_BLOCK_SIZE.

Structures

- Consists of two or more preallocated files.

(VI) Streams Pool:

- Introduced in Oracle 10g for Oracle Streams feature.
- Sized by STREAMS_POOL_SIZE parameter.
- Manage data and events sharing in distributed environments.

Types of buffer Pools in database cache buffers:

- DEFAULT:
- Standard database buffer cache used by all objects unless otherwise indicated.
 - Dynamic parameter DB_CACHE_SIZE is automatically sized if SGA_TARGET is set.
- KEEP:
- Used for frequently used objects to cache.
 - DB-KEEP_CACHE specifies size.
- RECYCLE:
- Used for objects that are less likely to access again.
 - DB-RECYCLE_CACHE_SIZE specifies the size.

PGA:

- Stands for Program Global Area.
- Also known as private global Area.
- Memory reserved for each user process connecting to an Oracle database.
- Allocated when a process is created.
- Deallocated when the process is terminated.
- Used by only one process.

Purpose:

- Memory area allocated and private for one process in Oracle.

Defn:

- A program global area is a memory region containing data and control information for a single process.
- Sometimes called a "Process global area."

Non-shared memory area:

- A PGA is non-shared memory area to which a process can write.
- One PGA is allocated for each session process.

Configurations

⇒ Depends on connection configuration.

- Shared Schemas
- Dedicated Schemas

Shared Schemas Configuration

- Multiple users share a connection, reducing memory usage on the server.
- Session information is held in the SGA instead of the PGA.

Dedicated Schemas Configuration

- Each user process has its own connection to the database.
- PGA contains session memory for this configuration.

Sort Area (Portions of memory in which Oracle sorts data are called sort areas)

- Utilized for sort, bitmap, merge sort or hash join operations.

PGA - AGGREGATE - TARGET parameters

- Along side WORKAREA_SIZE_POLICY, allows DBAs to specify a total size for all work areas, enabling Oracle to manage and allocate memory between user processes.

MEMORY_TARGET parameters

manages PGA and SGA memory collectively to optimize performance.

Total instance PGA

- Sum of memory used by all individual PGAs.

- Represented to as the total instance PGA or instance PGA.

⇒ The PGA is divided into:

1. Session Area
2. Private SQL Area
3. SQL work Area
4. Cursor Area

Session Area

- Stores session information of each user.

- Stores session variables, login information, session status, etc

Session information

- if the instance is running without the multithreaded server, the PGA also contains information about the user's session.

- if the instance is running in multithreaded server configuration, this session information is not in the PGA, but instead allocated in the SGA.

(3) Private SQL Area

⇒ The private SQL area contains data such as:

- bind variable values
 - query execution work areas
- ⇒ Each session that issues an SQL statement has a private SQL area.

Location of private SQL areas

- The location of a private SQL area depends on the type of connection established for a session.
- If a session is connected through a dedicated server, private SQL areas are located in the process's PGA.

- However, if a session is connected through a shared server, part of the private SQL area is kept in the SGA.

(2) Cursor area

- cursor: A cursor is a handle or name for a private SQL area - an area in memory that holds a passed statement and other information for processing.
- Stores the information of cursors.

⇒ The private SQL area of a cursor is itself divided into two areas whose lifetimes are different:

The persistent area

- This area contains bind variable values.
- it is freed only when the cursor is closed.

The runtime area

- Oracle database creates this area as the first step of an execute request.
- Query execution state information
- SQL work areas.

(4) SQL work Areas

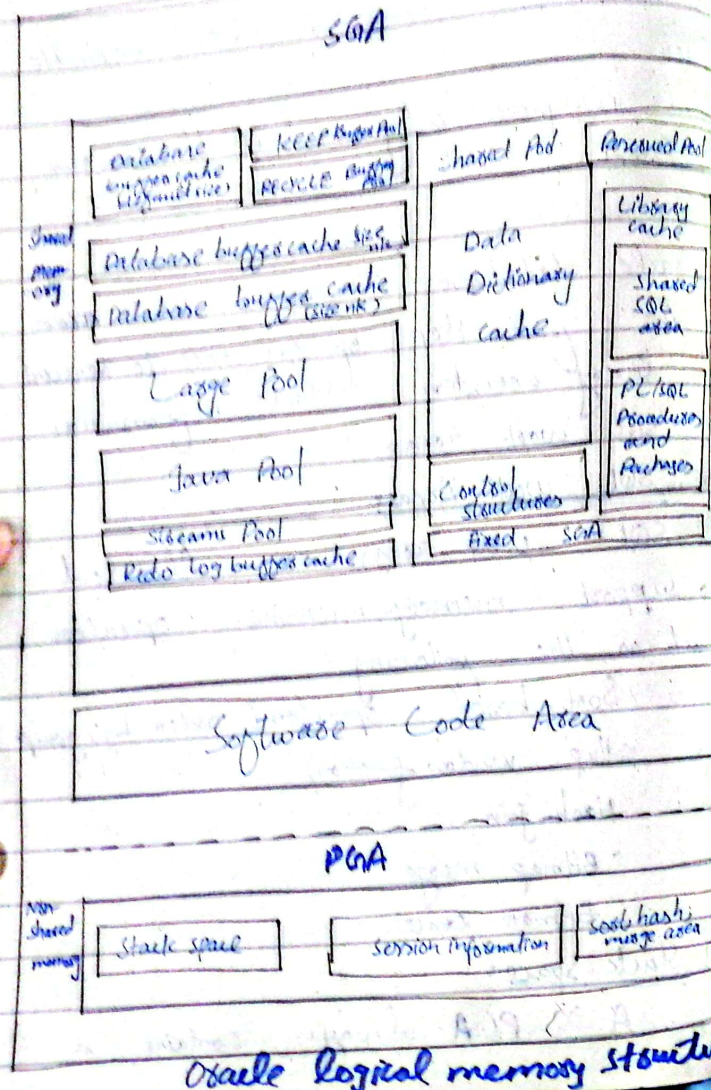
- SQL work areas are allocated to support memory-intensive operations such as the following:

- ⇒ Sort-based operators (order by, group by, rollup, window function)
- ⇒ Hash-join
- ⇒ Bitmap merge
- ⇒ Bitmap create

(5) Stack Space

- A PGA always contains a

stack space, which is memory allocated to hold a session's variables, arrays and other information.



Software Code Areas:

Software code areas are portions of memory used to store code that is being executed or may be executed.

Software areas are usually static in size, changing only when software is updated or reinstalled. The size of these areas varies by OS.

Stores Oracle executable files running within an Oracle instance.

Location:

- Typically resides in a privileged memory area separate from the user programs.

Read-only nature:

- Oracle software code is strictly read-only.

Installation options:

- Can be installed as Shared or non-shared.
- Shared installation saves memory when multiple Oracle instances are running on the same server at the same software release level.